

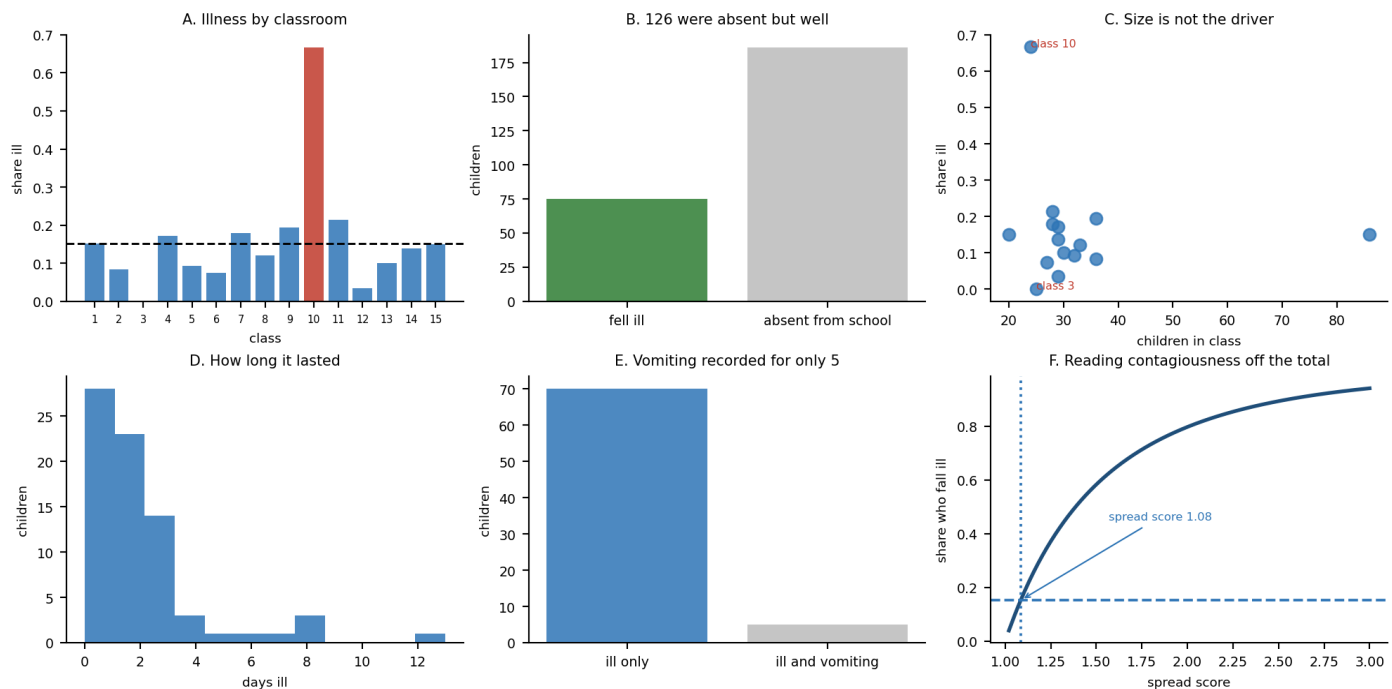
Norovirus — Derbyshire primary school, England, 2001

Dossier 4 of 4 · 492 children recorded individually · O'Neill & Marks (2005) Stat Med 24:2011

Findings, decisions, and where each one stands

Finding	Value in RAW data	Concern	Decision taken	Status in CANONICAL
Two candidate outcomes	75 ill vs 186 absent	126 children were absent but never recorded ill	count only children recorded unwell, never absence	RESOLVED — outcome is 'ill', 75 children
Classrooms differ sharply	chi-square 62.7, $p = 4e-08$	treating the school as one pool would be wrong	give each of the 15 classes its own adjustment	RESOLVED — clustering modelled explicitly
One extreme classroom	class 10: 16 of 24 ill (67%)	could single-handedly distort the estimate	kept in, absorbed by its class adjustment, not deleted	RESOLVED — rerun without it moves the answer by 0.009
One classroom with nobody ill	class 3: 0 of 25	a zero can break a naive calculation	kept in; its adjustment is pulled toward the school average	RESOLVED — no deletion, no imputation
Vomiting too rarely recorded	only 5 children ill and vomiting	too few to support any conclusion	excluded from the model	RESOLVED — excluded
No timeline	one record per child, not per day	cannot watch it spread over time	estimate contagiousness from the total share who fell ill	HANDLED — different method, same quantity

This shows that the important one is the first row: using school absence instead of recorded illness would have inflated this outbreak from 75 children to 186.



This shows that panel A is the reason each classroom gets its own adjustment. If the school were treated as one undifferentiated pool, class 10 would drag the estimate up and class 3 would drag it down, and neither effect belongs in the contagiousness figure.

What the model will estimate

Quantity	Pre-fit estimate	Published range	Reading
Spread score (R_0)	1.08	1.3 - 6.7 in institutions (Heijne 2012)	at the low end — a school day gives less contact than a care home
Share who fell ill	15.2% (75 of 492)	recorded directly	mild and self-limiting; most children recovered in two days
Usable observations	492 children	—	largest evidence base of the four, and the most precise result

This shows that because every child was recorded separately rather than as a daily tally, this outbreak supports the tightest estimate of the four.

VERDICT — CLEARED, and the strongest of the four. Six concerns, six decisions, all resolved. Individual-level records plus explicit classroom handling make this the most precise arm in the study.